A-9 Data Visualization II

1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names : 'sex' and 'age')
2. Write observations on the inference from the above statistics.

In [1]:
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
df=sns.load_dataset('titanic')
df.head()
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	decl
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN

In [3]:
```
df.tail()
```

Out[3]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	d
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	N
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	N
889	1	1	male	26.0	0	0	30.00	C	First	man	True	
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	N

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

In [5]:
```
df=df.dropna(subset=['age'])
```

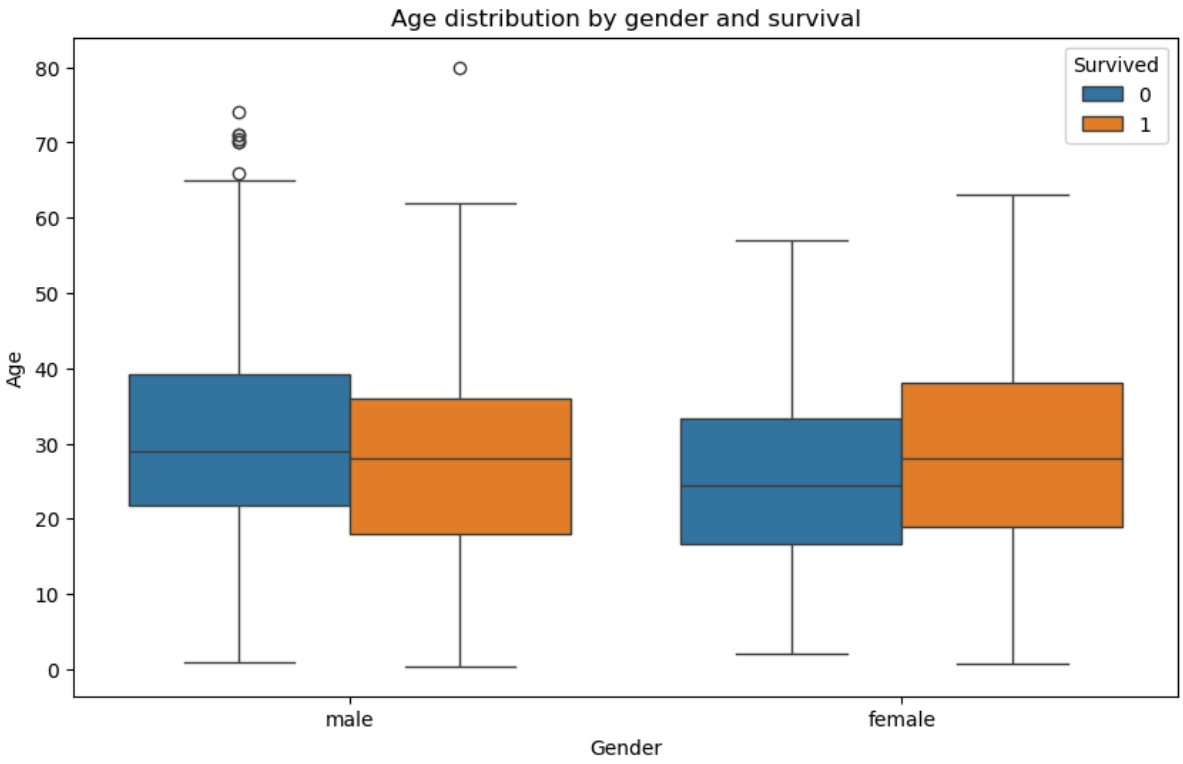
In [6]:
```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 714 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     714 non-null    int64
 1   pclass       714 non-null    int64
 2   sex          714 non-null    object
 3   age          714 non-null    float64
 4   sibsp        714 non-null    int64
 5   parch        714 non-null    int64
 6   fare         714 non-null    float64
 7   embarked     712 non-null    object
 8   class        714 non-null    category
 9   who          714 non-null    object
 10  adult_male   714 non-null    bool
 11  deck         184 non-null    category
 12  embark_town  712 non-null    object
 13  alive        714 non-null    object
 14  alone        714 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 70.2+ KB
```

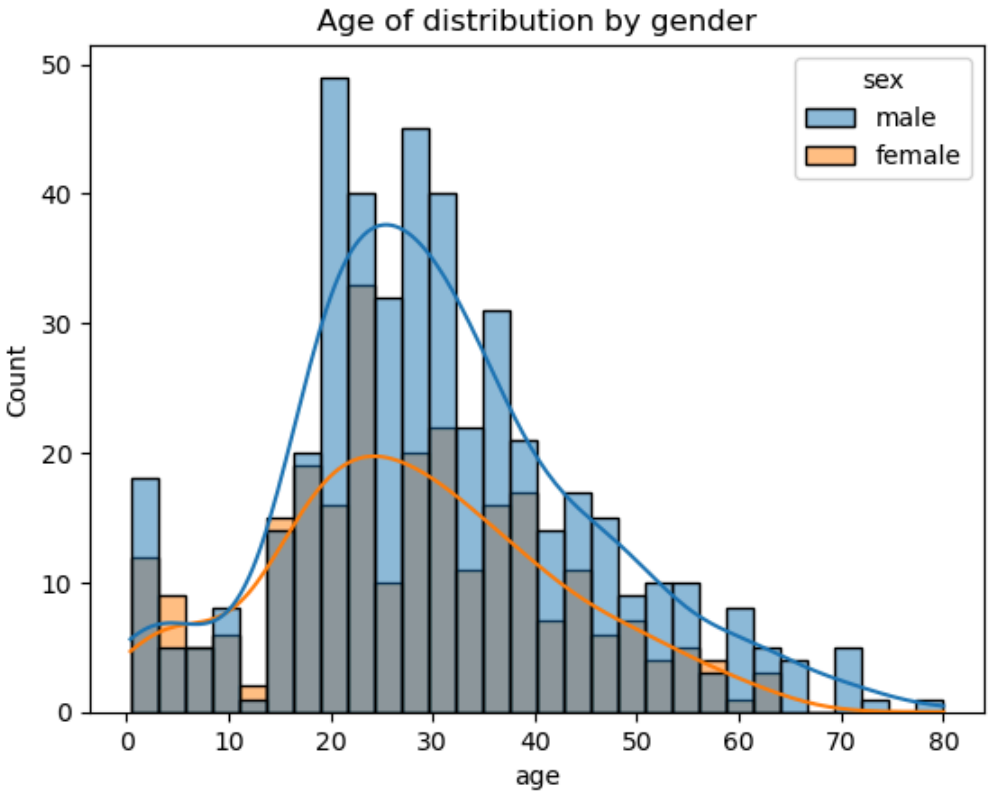
Out[6]:

	survived	pclass	age	sibsp	parch	fare
count	714.000000	714.000000	714.000000	714.000000	714.000000	714.000000
mean	0.406162	2.236695	29.699118	0.512605	0.431373	34.694514
std	0.491460	0.838250	14.526497	0.929783	0.853289	52.918930
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	1.000000	20.125000	0.000000	0.000000	8.050000
50%	0.000000	2.000000	28.000000	0.000000	0.000000	15.741700
75%	1.000000	3.000000	38.000000	1.000000	1.000000	33.375000
max	1.000000	3.000000	80.000000	5.000000	6.000000	512.329200

In [7]:
```
plt.figure(figsize=(10,6))
sns.boxplot(x='sex', y='age', hue='survived', data=df)
plt.title('Age distribution by gender and survival')
plt.xlabel('Gender')
plt.ylabel('Age')
plt.legend(title='Survived')
plt.show()
```



In [8]:
```
sns.histplot(data=df, x='age', hue='sex', bins=30, kde=True)
plt.title('Age of distribution by gender')
plt.show()
```



In [9]:
```
sns.violinplot(x='sex', y='age', hue='survived', data=df, split=True)
plt.title('Violin plot of age distribution')
plt.show()
```

Observation and inference : -> Female passengers show higher survival rates compared to males -> Median age of survivors is slightly lower than non survivors (younger people have better survival chances) -> Several outliers exist in both male and female categories (elder passengers with lower survival rates)